

E-Commerce Payment Gateway - Non-Functional Requirements

Confidential - Internal Use Only

Version 1.0

1. Performance Requirements

- NFR-ECO-001: Payment API latency must be < 500ms at P95 for end-to-end payment processing.
- NFR-ECO-002: Hosted checkout page must achieve Largest Contentful Paint (LCP) < 2.0 seconds on 3G connections.
- NFR-ECO-003: Merchant dashboard must load within 3 seconds with up to 10,000 transactions in view.
- NFR-ECO-004: Webhook delivery must complete within 5 seconds of event occurrence for 99% of events.

2. Scalability Requirements

- NFR-ECO-005: System must handle 10,000 transactions per minute at peak load without degradation.
- NFR-ECO-006: Horizontal auto-scaling must trigger when CPU > 70% sustained for 2 minutes. Scale-in at < 30% for 10 minutes.
- NFR-ECO-007: Database read replicas must handle 80% of read traffic; write primary must handle 2,000 writes/sec.
- NFR-ECO-008: Kafka cluster must sustain 50,000 events/second with 3-day retention.

3. Availability & Disaster Recovery

- NFR-ECO-009: System uptime SLA: 99.99% (max 4.38 minutes downtime per month).
- NFR-ECO-010: Multi-AZ deployment with automatic failover. RTO < 5 minutes, RPO < 1 minute.
- NFR-ECO-011: Cross-region disaster recovery in US-West-2 (primary) and EU-West-1 (DR). Monthly DR drill mandatory.
- NFR-ECO-012: Circuit breaker pattern on all external service calls (card networks, XE API) with fallback to cached data.

4. Security Requirements

- NFR-ECO-013: All inter-service communication must use mutual TLS (mTLS) with certificate rotation every 90 days.
- NFR-ECO-014: Secrets management via AWS Secrets Manager with automatic rotation for DB credentials and API keys.
- NFR-ECO-015: OWASP Top 10 vulnerability scan must pass with zero critical/high findings before each release.
- NFR-ECO-016: Container images must be scanned with Trivy; no critical CVEs allowed in production images.

5. Accessibility & Internationalization

- NFR-ECO-017: Hosted checkout must comply with WCAG 2.1 Level AA. All form fields must have associated labels and aria attributes.
- NFR-ECO-018: Checkout page must support RTL layout for Arabic and Hebrew locales.
- NFR-ECO-019: Error messages must be localizable. Default language: English. Supported: EN, ES, FR, DE, JA, ZH, AR.
- NFR-ECO-020: Keyboard navigation must allow full checkout completion without mouse interaction. Tab order follows visual top-to-bottom flow.

6. Observability & Monitoring

NFR-ECO-021: Structured JSON logging with `correlation_id` propagated across all services.

NFR-ECO-022: Distributed tracing via OpenTelemetry with Jaeger for visualization. Trace sampling rate: 10% normal, 100% on error.

NFR-ECO-023: Prometheus metrics exported for: `request_count`, `request_duration_seconds`, `error_rate`, `active_connections`.

NFR-ECO-024: PagerDuty alerts for: error rate > 5%, P95 latency > 1s, settlement batch failure, fraud score anomaly.